DetectX

ANANTAPUR POLICE DEPARTMENT & INCUXAI

AUTO CHALLAN SYSTEM

Smart Surveillance for Safer Roads

PROJECT DETAILS

DetectX

Team Code: INC-APT-2026-192

Team Leader

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PROBLEM STATEMENT

PROBLEM

- Every day in India, thousands of people ride motorcycles without helmets and carry 3+ people on a single bike. Traffic police can't be everywhere they're limited in number, can only monitor a few junctions at a time, and manually spotting + chasing down violators is slow and inconsistent.

- The result? Most violations go unpunished. People know the chances of getting caught are low, so they keep doing it. And the death toll keeps rising **over 1.5 lakh people die in road accidents every year in India**, with two-wheelers accounting for the majority.

The Core Gap

- Existing CCTV cameras at traffic signals are mostly just recording. Nobody is watching all of them in real time. There aren't enough officers to review footage manually.

PROBLEM STATEMENT

SOLUTION

- **INTEGRATION OF AI** — Integrating AI to auto detect violations
- **SPOTS THE VIOLATION** — No Helmet, Triple Riding
- **AUTO-GENERATE THE CHALLAN** — With photo evidence, timestamp, fine amount
- **LOGS DATA** — Ready to export and act on

FILE STRUCTURE & TECH STACK

PROJECT FILES

```
DetectX/

  ├── detectx_server.py   Flask backend + AI logic

  ├── requirements.txt   Python dependencies

  ├── detectx_dashboard.html   Main UI page

  ├── detect_dashboard.css   All styles

  └── detect_dashboard.js   Frontend logic
```

TECH STACK

Python / Flask
Backend API server, routing, CORS

AI Models
OpenCV and Mediapipe model used to analyze the image and detect triple riding or no helmet

OCR & YOLO
OCR and ultralytics(YOLO) for fast detection of objects from live stream and OCR for number plate detect form the object

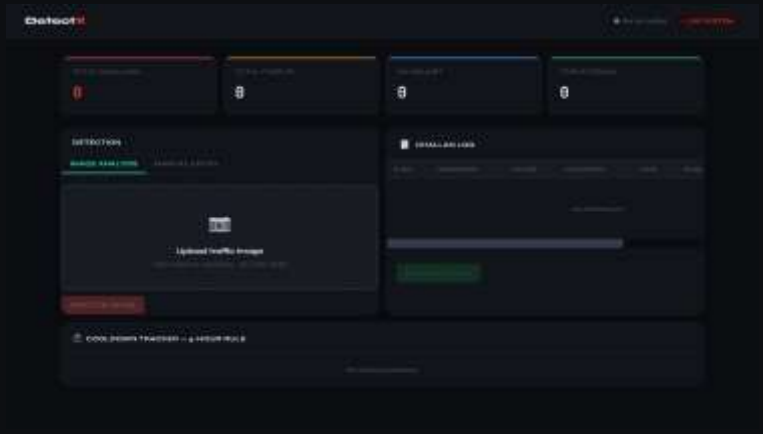
HTML + CSS + JS
Frontend UI, tabs, upload, challan log

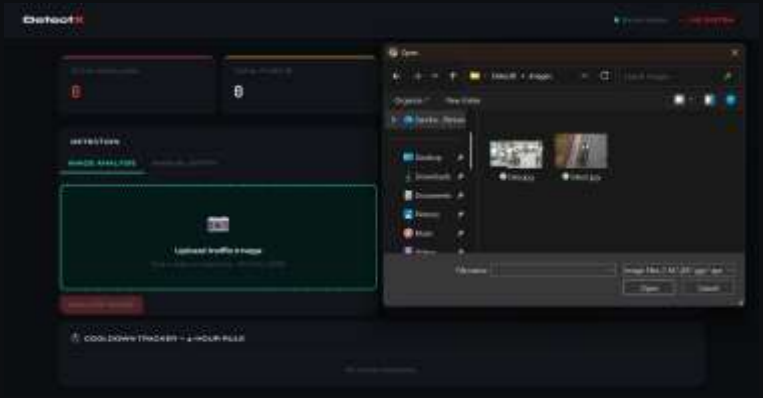
SheetJS (xlsx)
Client-side Excel export with 2 sheets

Canvas API
Image normalization → JPEG before API call

PAGE 1 · detectx_dashboard.html — Main UI

FRONTEND





Header & Server Status

Sticky header with live Flask server indicator — green dot = online, red = offline. Re-checks every 30 seconds automatically.

4 Stats Cards

Live counters: Total Challans, Total Fines (₹), No Helmet count, and Overloaded count. Updates instantly after each challan.

Detection Card (Left)

Two tabs: AI Image Analysis (upload → Flask → AI Logic) and Manual Entry (checkbox violations + plate input).

Challan Log Card (Right)

Auto-populating table with S.No, plate badge, violation pill, fine, challan ID, and time. One-click Excel export.

Cooldown Tracker

Spans full width. Shows all vehicles under 4-hour cooldown with a live countdown timer per plate.

PAGE 2 · detectx_erver.py — Flask Backend

PYTHON

detectx_server.py

```python
from flask import Flask, request, jsonify, send_from_directory, after_this_request
import cv2, numpy as np, base64, os, json, pandas as pd
from ultralytics import YOLO
import mediapipe as mp
import easyocr
from datetime import datetime
from openpyxl import load_workbook
from openpyxl.drawing.image import Image as XLImage
from openpyxl.utils import get_column_letter

EXCEL_FILE  = "detectx_logs.xlsx"
SNAP_DIR    = "violation_snapshots"
YOLO_MODEL  = "yolov8n.pt"
FINE_TRIPLE = 1000
FINE_HELMET = 1000

os.makedirs(SNAP_DIR, exist_ok=True)

app = Flask(__name__, static_folder=".", static_url_path="")

@app.after_request
def add_cors(response):
    response.headers["Access-Control-Allow-Origin"]  = "*"
    response.headers["Access-Control-Allow-Headers"] = "Content-Type"
    response.headers["Access-Control-Allow-Methods"] = "GET, POST, OPTIONS"
    return response

@app.route("/analyze", methods=["OPTIONS"])
@app.route("/ping",    methods=["OPTIONS"])
def options_handler():
    return "", 204
```

GET /health

Health Check Endpoint

The frontend polls this every 30s to drive the server status indicator in the header.

POST /analyze

AI Analysis Endpoint

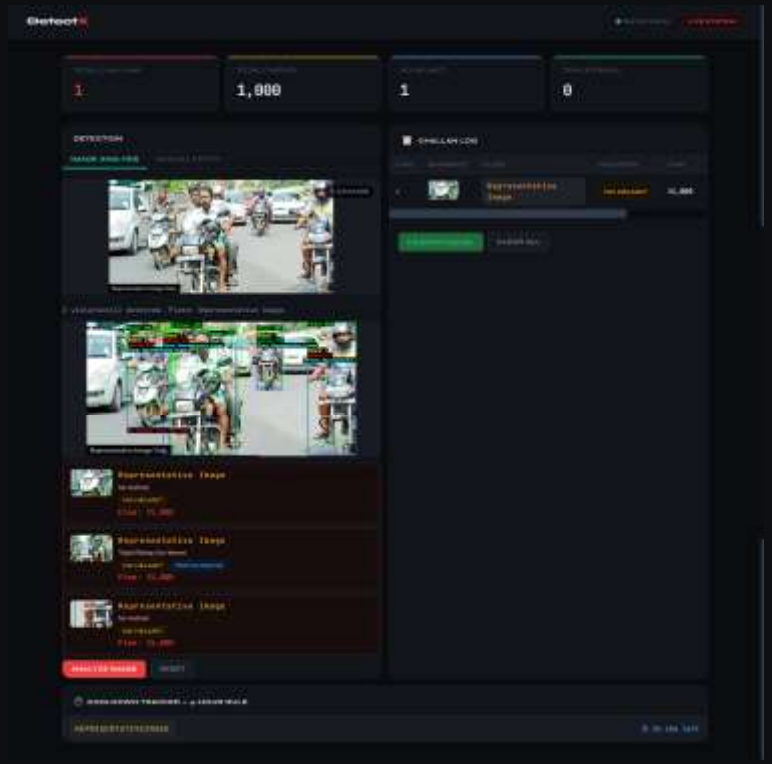
Fine Calculation Logic (Python)

Rider has no helmet	₹ 1,000
More than 2 persons on bike	₹ 1,000
No helmet AND more than 2 persons	₹ 2,000
Same plate within 4 hours	BLOCKED

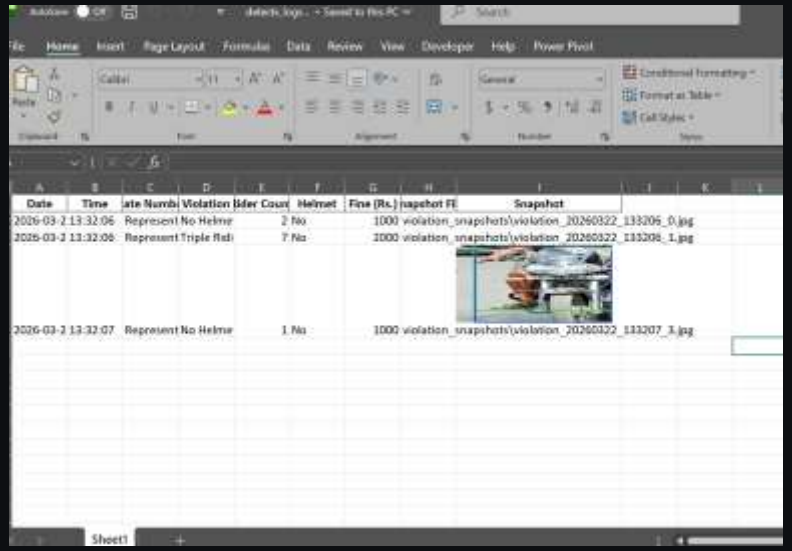
PAGE 3 · Output page

RESULT

WEB OUTPUT INTERFACE



DATA LOGS IN EXCEL



Date	Time	ate Numb	Violation	ider Coun	Helmet	Fine (Rs.)	napshot FE	Snapshot
2026-03-2	13:32:06	Represent	No Helme	2	No	1000	violation_snapshots\violation_20260322_133206_0.jpg	
2026-03-2	13:32:06	Represent	Triple Rid	7	No	1000	violation_snapshots\violation_20260322_133206_1.jpg	
2026-03-2	13:32:07	Represent	No Helme	1	No	1000	violation_snapshots\violation_20260322_133207_3.jpg	

Sheet1

PAGE 4 · Violation Logs with Snapshots

Date	Time	Plate Number	Violation	Rider Count	Helmet	Fine (Rs.)	Snapshot File	Snapshot
2026-03-22	13:32:06	XX XX XX XXXX	No Helmet	2	No	1000	violation_1.jpg	
2026-03-22	13:32:06	XX XX XX XXXX	Triple Riding, No Helmet	3	No	2000	violation_2.jpg	
2026-03-22	13:32:07	Not Detected	No Helmet	1	No	1000	violation_3.jpg	

Embedded Snapshot
Image inside Excel cell

Auto Appends
Each run adds new rows

Lock-Safe Write
Timestamped backup if locked

Browser Export
One-click .xlsx from dashboard

PREVENTING DUPLICATE FINES

How Cooldown Works

The Problem it Solves

The same vehicle could appear in multiple frames or images and get fined repeatedly for the same violation. The 4-hour cooldown prevents this.

How It Works

→ Challan issued for plate TN09BT9721
→ Timestamp stored in cooldowns{}
→ Same plate detected again in 2 hrs
→ **BLOCKED — challan not re-issued**
→ After 4 hours — plate is eligible again

Manual Entry Warning

When typing a plate in Manual Entry, if cooldown is active the remaining time is shown and the Generate button is disabled.

Active Cooldowns

TN09BT9721	3h 28m left
TN02AV6400	3h 28m left
MH12AB1234	1h 45m left

0h ———————————— 4h
1h 28m elapsed — 2h 32m remaining

Auto-Refresh

Tracker updates every 30 seconds automatically. Expired entries disappear without refresh.

Fair Enforcement

No vehicle gets fined more than once per 4-hour window, even across multiple uploaded images.

REFERENCES

Image - 1

Image Credit: Rayomand Engineer
Website: The Better India
Posted Date: 24-July, 2018
Link : https://thebetterindia.com/152186/iit-hyderabad-ai-module-helmet-motorcycle-news/

Image - 2

Stock Photo ID: #EA595702
Image Credit: Picxy.com/PraveenPulletikurru
Model Release: NA
Link : https://www.picxy.com/photo/271392

DetectX

Auto Challan System · Smart Surveillance for Safer Roads

No Helmet | Auto Challan | Excel Logs

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Intelligent · Automated · Accurate